

Numerical Methods for the Radiative Transfer Equation

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1 Formal Solution of the Radiative Transfer Equation

Consider the radiative transfer equation along a fixed direction in a plane-parallel medium. Along a ray parameterized by path length s , the equation may be written as

$$\frac{dI(s)}{ds} = -\kappa(s)I(s) + j(s). \quad (1)$$

This is a first-order linear ordinary differential equation. We rewrite it in standard linear form:

$$\frac{dI}{ds} + \kappa(s)I = j(s). \quad (2)$$

The integrating factor is

$$\mu(s) = \exp \left(\int_0^s \kappa(s') ds' \right). \quad (3)$$

Multiplying through:

$$\mu \frac{dI}{ds} + \mu \kappa I = \mu j. \quad (4)$$

The left-hand side becomes a total derivative:

$$\frac{d}{ds} [\mu(s)I(s)] = \mu(s)j(s). \quad (5)$$

Integrating from 0 to s :

$$\mu(s)I(s) - I(0) = \int_0^s \mu(s')j(s') ds'. \quad (6)$$

Solving for $I(s)$:

$$I(s) = e^{-\int_0^s \kappa(u) du} \left[I(0) + \int_0^s e^{\int_0^{s'} \kappa(u) du} j(s') ds' \right]. \quad (7)$$